INSTITUTE OF SONOLOGY

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Synthesis

Composing with Algorithms

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Synthesis

Jean-Claude Risset: Synthesis is primarily a psychoacoustic problem: what are the appropriate parameters for the imagined sound needed?

J.K Randall: Sound synthesis is not a scientific problem but a compositional one: "Musical composition continues to play the central role in defining the current frontiers of the aurally perceptible."

Koenig: Digital synthesis do not need ‘referring to a given acoustic model but rather describing the waveform [directly] in terms of amplitude values and time values’ this avoids the goal of emulating, modelling or imitating any other instrumental or electronic paradigm.

Synthesis

Synthesizing sound with signals and the transformation of those signals to create musical material.

Techniques discussed:

* *Additive Synthesis*
* *Noise-based Synthesis*
* *Subtractive Synthesis*
* *Modulation Synthesis*
* *Physical Modelling*

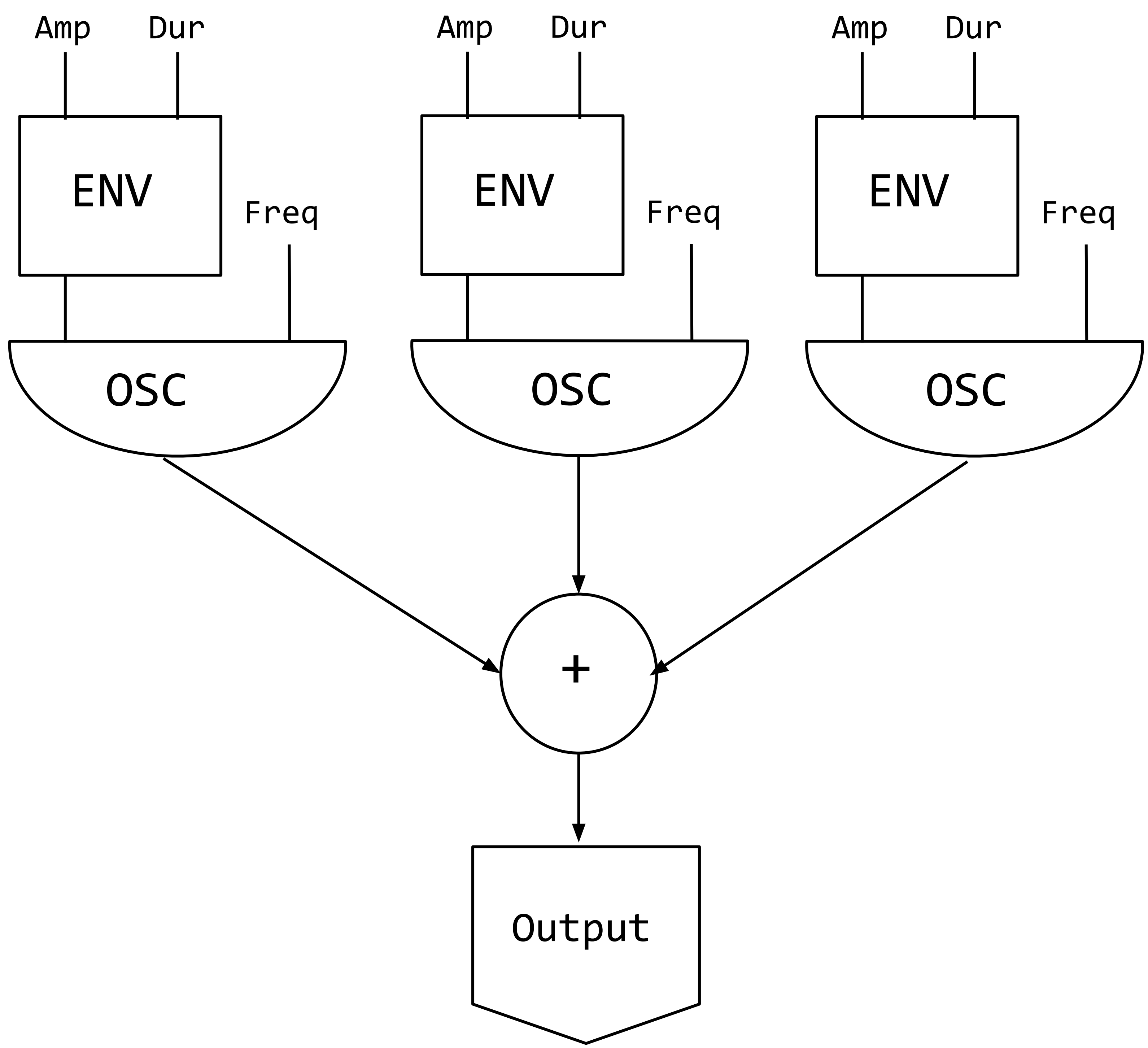
Additive synthesis

Additive synthesis is a sound synthesis technique that creates timbre by **adding** sine waves together.

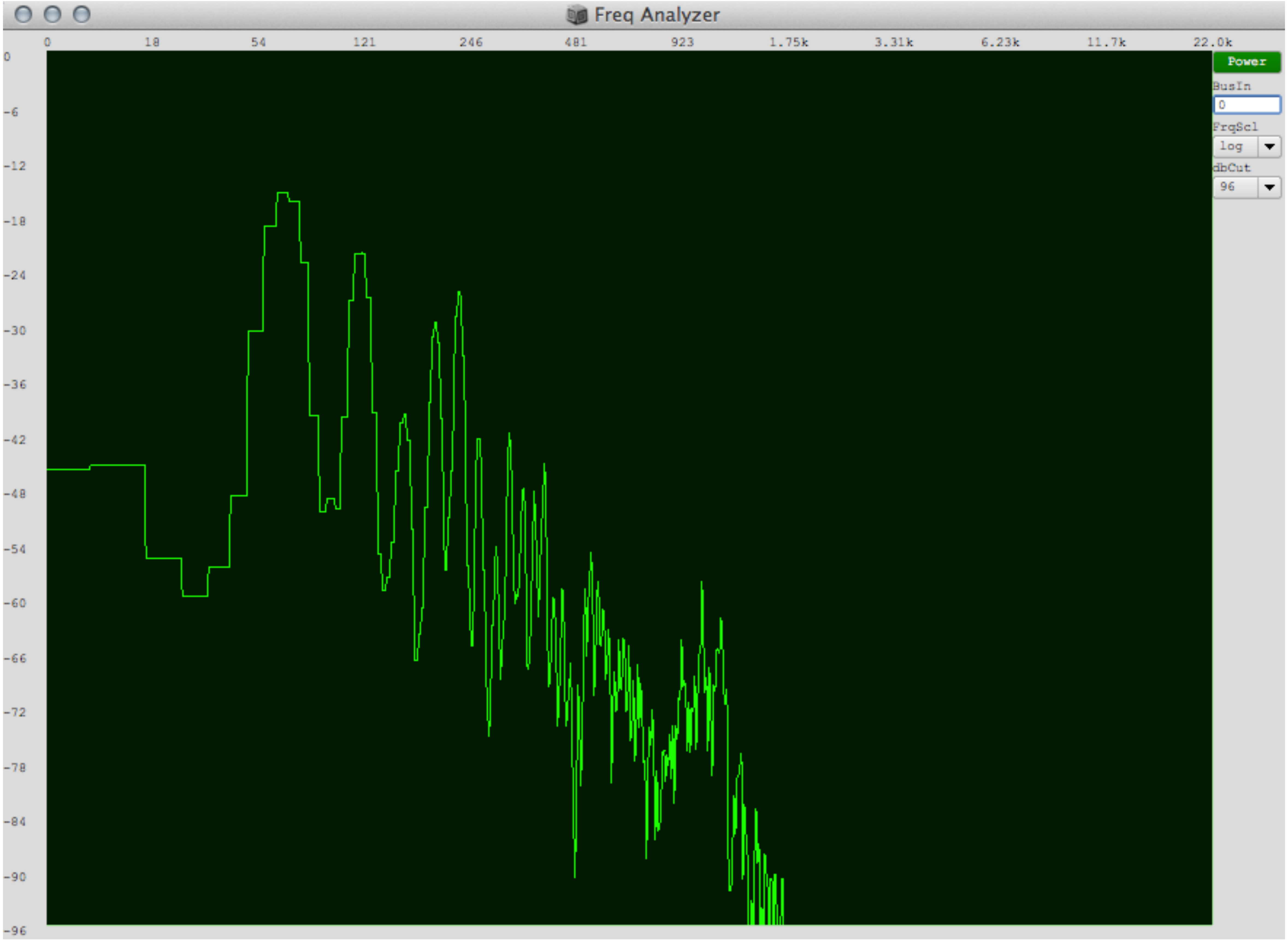
According to Fourier theory any periodic waveform can be broken down into **a series of sine waves at different frequencies, amplitudes, and phases.**

The layering of sinusoidal waves and building sounds from individual partials requires many oscillators and large data specification in order to obtain complex sounds.

Additive synthesis



Frequency Spectrum



Cologne

The Cologne studio in the 1950s made much use of its sine wave oscillator. Additive synthesis was done by summing the partials on tape after recording them.

By using the **sine wave** as a starting point serial methods could be realized where *"everything to the last element of the note is subjected to serial permutation"* (Eimert, 1955)

This extremely laborious and time-consuming activity resulted in various developments in electronic music and for composing sound.

Noises

The **color** of a **noise** signal (a signal produced by a stochastic process) is generally understood to be some broad characteristic of its power spectrum.

This sense of "color" for noise signals is similar to the concept of timbre in music.

Noise

Noises are usually named after **colors**. This started with *"white noise"*, a signal whose spectrum has equal power within any equal interval of frequencies. That name was given by analogy with *"white light"*, which was (incorrectly) assumed to have such a *"flat"* power spectrum over the visible range.

Other color names, like *"pink", "red", and "blue"* were then given to noise with other spectral profiles; often (but not always) in reference to the color of light with similar spectra.

Filters

It is common to use filters during synthesis to shape a signal. This works best when the source being filtered has a rich spectrum.

Low Pass – allows all frequencies below the cutoff point to pass through.

High Pass – the opposite of the low pass filter, allows all frequencies above the cutoff point to pass through.

Band Pass – allows a band on frequencies to pass through in the centre, but stops all frequencies outside of this band.

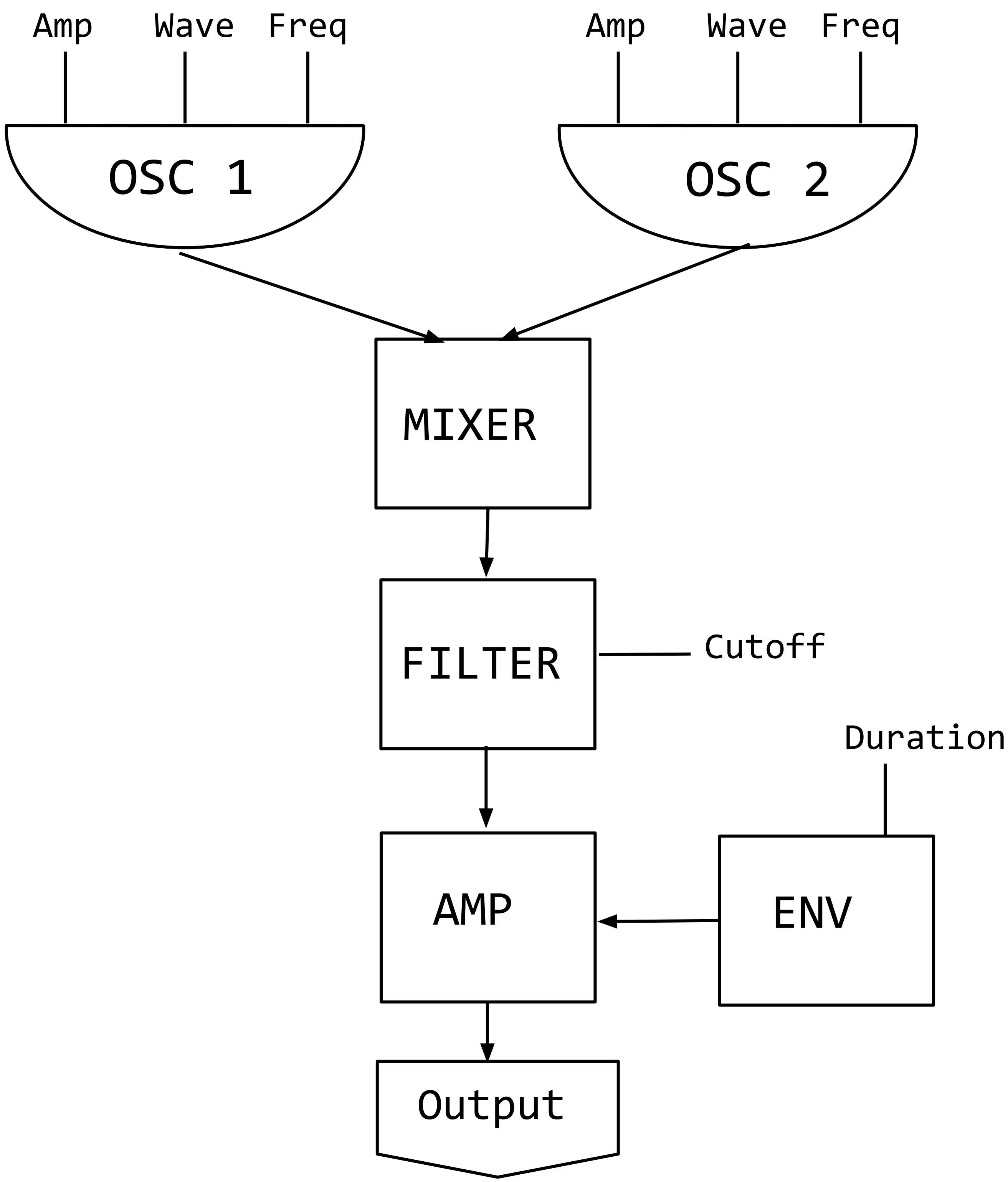
Band Notch/Reject – stops a band of frequencies in the centre from passing through.

Subtractive Synthesis

A method of sound synthesis where partials of an audio signal (often one rich in harmonics) are a**ttenuated by a filter** to alter the timbre of the sound.

Can be applied to any source audio signal, the sound most commonly associated with the technique is that of analog synthesizers of the 1960s and 1970s, in which the harmonics of simple waveforms such as sawtooth, pulse or square waves are attenuated with a voltage-controlled resonant low-pass filter.

Subtractive synthesis



Ring modulation

A signal processing method that multiplies two signals, a **carrier** (C) and a **modulator** (M).

When both are sine waves the resulting spectrum contains two frequencies, one at **C+M** and the other at **C -M**.

For richer sounds many more sidebands occur. RM can be used to make a sound 'richer' or appear more 'electronic'.

Amplitude modulation

Related to RM and is also uses a **carrier** wave modulated by a **modulator** one.

The difference is that AM uses a **unipolar** modulator which means it never goes below 0.

Tremolo effects in electronic music are often based on AM.

FM Synthesis

Discovered by John Chowning in 1967.

Frequency Modulation Synthesis (or FM synthesis) is a synthesis type where the timbre of a simple waveform (such as a square, triangle, or sawtooth) is altered by modulating its frequency with a modulation waveform that is also in the audio range

FM sounds ca result in a more complex waveforms and generate a great variety of different sounds with economic means.

FM Synthesis

If the **modulator** frequency in FM synthesis is lower than 30 Hz the modulation will sound like a vibrato. If the frequency is higher new frequencies (sidebands) appear in the signal.

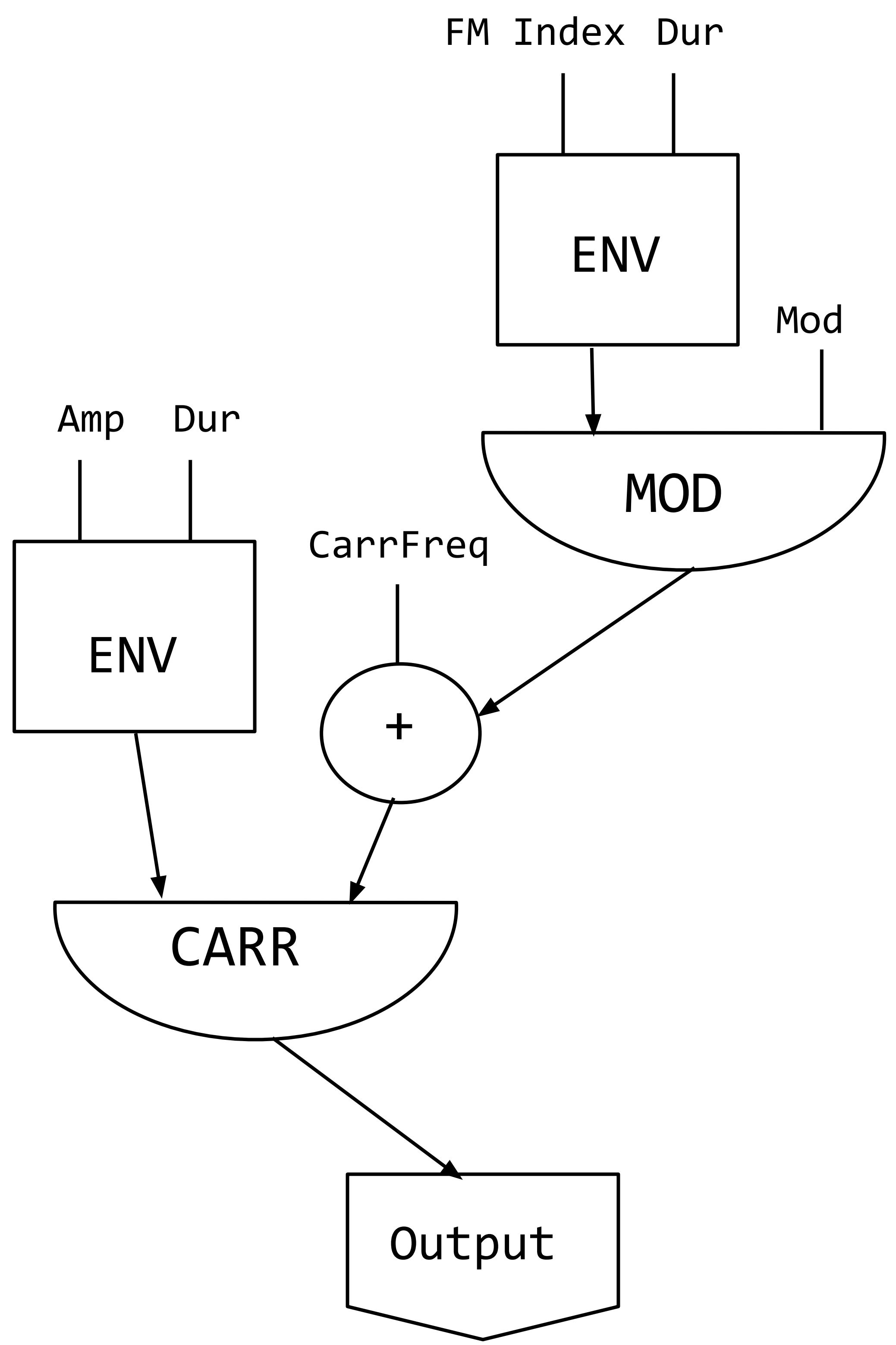
The frequencies of the **sidebands** depend on the **ratio** of the carrier frequency to the modulation frequency. This ratio is called the **C : M Ratio**. If the ratio is a simple integer, <u>harmonic</u> spectra is produced otherwise <u>inharmonic</u> spectra will result.

FM Synthesis

The **strength** of the **sidebands** (bandwidth and brightness of spectrum) is controlled by the **modulation index** I.

I is defined as **I = D / M** where **D** is the **depth** of **modulation** (measured in deviation of frequency) and **M** is the **modulation frequency**.

FM synthesis



Physical Modelling

Methods in which the waveform of the sound to be generated is computed by using a **mathematical model** to **simulate** a **physical source** of sound, usually a musical instrument.

Such a model is usually made of laws obtained from physics that control the sound production and will have several parameters a user will focus on.

The model often includes constants (for describing spaces or physical materials) but also time-dependent functions modeling the interaction with the external world (how an instrument is played).

Physical Modelling

In many cases PM techniques can be split into **excitation** and **resonance models**.

Excitation is mostly generated by noise-sources while the **resonances** use delay-lines.

A well known example is the Karplus-Strong method where a short noise burst is fed through a filtered delay line to simulate the sound of a hammered or plucked string or percussion instrument.

Exercises

```
(
    NF(\iop, {|freq=78, mul=1.0, add=0.0|
        var noise = LFNoise1.ar(0.001).range(freq, freq + (freq * 0.1));
        var osc = SinOsc.ar([noise, noise * 1.04, noise * 1.02, noise * 1.08],0,0.2);
        var out = DFM1.ar(osc,freq*4,SinOsc.kr(0.01).range(0.92,1.05),1,0,0.005,0.7);
        HPF.ar(out, 40)
    }).play;
)

(
    NF(\dsc, {|freq = 1080|
        HPF.ar(
            BBandStop.ar(Saw.ar(LFNoise1.ar([19,12]).range(freq,freq*2), 0.2).excess(
            SinOsc.ar( [freq + 6, freq + 4, freq + 2, freq + 8])),
            LFNoise1.ar([12,14,10]).range(100,900),
            SinOsc.ar(20).range(9,11)
        ), 80)
        ;
    }).play;
)
```

```
var <>pindex, <>cindex;

initialize {
    if(pindex.isNil, { pindex = 1000 });
    if(cindex.isNil, { cindex = 2000 });
}

clearProcessSlots {
    pindex = 1000;
    (this.pindex - 1000).do{|i| this[this.pindex+i] = nil; }
}

clearOrInit {|clear=true|
    if(clear == true, { this.clearProcessSlots() }, { this.initialize() });
}

transform {|process, index|
    if(index.isNil && pindex.isNil, {
        this.initialize();
    });

    pindex = pindex + 1;
    this[pindex] = \filter -> process;
}

control {|process, index|
    var i = index;

    if(i.isNil, {
        this.initialize();
        cindex = cindex + 1;
        i = cindex;
    });

    this[i] = \pset -> process;
}
```

```
(
    NF(\depfm, {|freqMin=5, freqMax=20, mul=20, add=80, rate=0.5, modFreq=2100, index=0.3, amp=0.2|
        var trig, seq, freq;
        trig = Dust.kr(rate);
        seq = Dwhite(freqMin, freqMax, inf).midicps;
        freq = Demand.kr(trig, 0, seq);
        HPF.ar(PMOsc.ar(LFCub.kr([freq, freq/2, freq/3, freq/4], 0, mul, add),
        LFNoise1.ar(0.3).range(modFreq,modFreq*2), index) * amp, 50)
    }).play;
)
```

Exercises

1. Create a sound movement that contains a sine wave and a noise.

2. Create a cluster of sine wave (10 or more) at random frequencies.

3. Create a Subtractive synthesis with a saw wav and a lowpass filter.

4. Combine different moments of noise types where each is filtered in a different way.

5. Create a FM sound using two carrier waves each one being modulated by a different modulator.